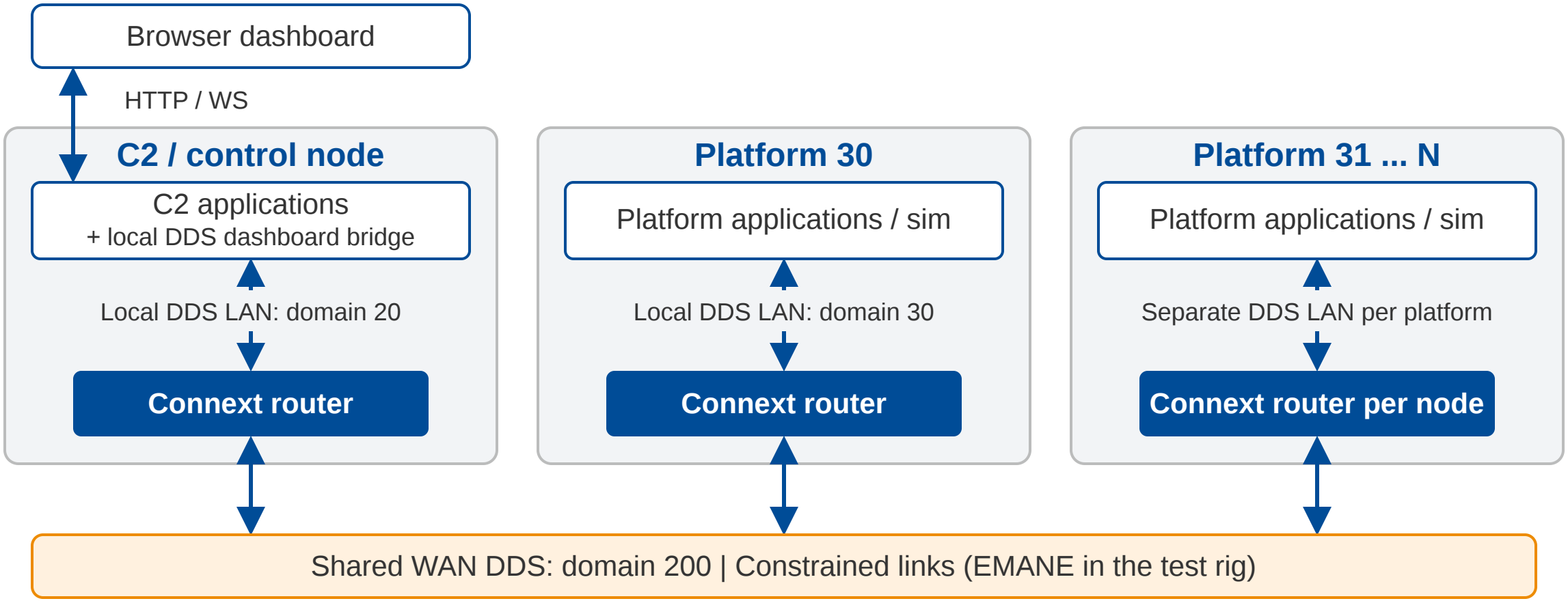
Current system at a glance



One router per node: selected commands, status, health, and team data cross the WAN.

Isolation mechanisms at a glance

Mechanism	Network	Discovery	Logical	Runtime changeable
VLANs / firewall rules	Network paths	—	—	Network admin
Domain ID	Port ranges	Participant + endpoint	—	—
Domain tag	—	Participant + endpoint	—	—
DomainParticipant partitions	—	Participant + endpoint	Subsystem / team	Application
Publisher/Subscriber partitions	—	—	Groups of topics	Application
Topic names	—	—	Topic	—
Content filters	—	—	Sample	Application
Union discriminator	—	—	Sample (variant)	Application

Runtime changeable: no DDS entity recreation.

Discovery options at a glance

Option	How peers are found	Benefit	Trade-off
Multicast	Announce to a shared multicast group	No peer address roster	Requires multicast-capable network
Unicast: defined peers	Contact configured discovery addresses	No multicast or central service required	Maintain peer addresses as deployment changes
CDS	Cloud Discovery Service forwards participant announcements	Dynamic rendezvous without a custom directory	Operate a reachable discovery service
Unicast: bootstrap	A directory (e.g. C2) supplies peers; application adds them at runtime	Start with a bootstrap address, not the full roster	Application owns directory updates and recovery

Bootstrap: knowing one ordinary participant does not automatically discover its peers.

DDIL link-health measurement options

Measurement	Simple approach	What it tells us
Peer presence	Track existing health heartbeats	Is the peer reachable?
Round-trip time (RTT)	Acknowledge a heartbeat in the next heartbeat; subtract the reporting wait	How long is the round trip?
DDS protocol counters	Track per-peer repairs and unacknowledged data	Is reliable delivery under pressure?

Reuse existing traffic. Show how fresh each measurement is.

Presence, delay, and delivery pressure are separate facts, not a single health score.